

Zhenghe Guo

3230100923@zju.edu.cn
styokuok.github.io/main GitHub Google Scholar

Final-year undergraduate researcher working on embodied systems and spatial intelligence, with a focus on learned state representations, world modeling, and planning.

Education

Zhejiang University

B.Eng. in Agricultural Engineering; Minor in Advanced Class of Engineering Education
GPA: 3.86/4.00. Undergraduate research with Prof. Mingchuan Zhou.

Hangzhou, China

2023–2027

National University of Singapore

Visiting Student, Statistics and Data Science (Global Science Summer Programme)
GPA: 4.00/4.00; decision-tree methods instructed by Prof. Wei-Yin Loh.

Singapore

Jun–Jul 2025

Research Experience

Westlake University

Visiting Research Intern, Inception3D, led by Prof. Anpei Chen

Research on deformable-object manipulation and physics-grounded world models, combining MPM-based simulation with learned residual dynamics.

Hangzhou, China

May–Oct 2026

Tsinghua University, Institute for AI Industry Research (AIR)

Visiting Research Intern, DISCOVER Lab

Integrated a StereoHO-based perception baseline, calibration, visualization, and robot-side infrastructure for stereo hand-object reconstruction and handover experiments.

Beijing, China

Jan–Feb 2026

Zhejiang University, Robotic Micro/Nano Manipulation Lab

Undergraduate Research Assistant, advised by Prof. Mingchuan Zhou

Developing modeling, simulation, kinematics, and autonomous navigation methods for a magnetically controlled zebrafish robot.

Hangzhou, China

Feb 2025–present

Zhejiang University, State Key Laboratory of CAD&CG

Undergraduate Research Assistant, advised by Prof. Ruizhen Hu

Research on instance-level reconstruction, with additional work on physics-grounded articulated-object reconstruction and street-scene reconstruction using COLMAP and FreeTimeGS.

Hangzhou, China

Oct 2025–Jun 2026

Zhejiang University, Agile Robotic Tele-systems (ARTs) Lab

Research Assistant, advised by Prof. Gaofeng Li

Research on tactile sensing for biomimetic soft-tissue stress-field reconstruction, including sensor-layout optimization, active and sparse sampling, Gaussian-process inference, curved-surface tactile sensing, simulation, and data processing.

Hangzhou, China

Oct 2024–present

Publications

Zhenghe Guo. “ObjectTwin: Measuring the Asset Correspondence Gap for Capture-Anchored Digital Twins.” Project page

X. Yang, **Z. Guo**, S. Liang, Z. Yang, C. Zhu, and Z. Zhang. “Rate Maximization for UAV-assisted ISAC System with Fluid Antennas.” *2025 IEEE/CIC International Conference on Communications in China Workshops*, pp. 1–5, 2025. doi:10.1109/ICCCWorkshops67136.2025.11148178

Zhenghe Guo et al. “Action Identification and Validation Design for Prediction with Missing Covariates.” Manuscript in preparation.

Selected Projects

ZJUDancer RoboCup Humanoid Vision and Localization

2025–2026

- Developed and debugged ball, field, and robot perception, self-localization, and odometry integration for autonomous humanoid soccer; First Prize and Runner-up at RoboCup China Open 2026.

DYNAMO – Unitree G1 Dance Motion Planning

2025

- Designed a dynamics-aware framework combining contact scheduling, ZMP-aware centroidal control, whole-body coordination, and electrical-energy analysis; Asia and Pacific Mathematical Contest in Modeling (APMCM) National First Prize (top 2%).

Selected Projects (continued)

RoboARM – Vision-Guided Gomoku Manipulation 2025

- Integrated YOLO and ArUco perception, an HTTP-to-ROS decision bridge, Negamax search, calibration, CAN-based six-axis arm control, and a pneumatic end effector into an autonomous pick-and-place loop.

TurtleBot Autonomous Navigation 2025

- Implemented a ROS RRT* global planner with goal-biased sampling, KD-tree collision queries, obstacle inflation, parent selection, rewiring, path publication, and real-robot visualization.

Hybrid QUBO LDPC Decoder 2025

- Engineered an LDPC simulation and decoding prototype with belief-propagation baselines, QUBO construction, adaptive penalties, partitioned solving, and BER/FER/runtime evaluation.

Selected Awards

Robotics

- RoboCup China Open, Humanoid League (Small) – First Prize, Runner-up, Team ZJUDancer, 2026.
- Zhongkong Cup Robotics Competition – Second Prize, 2025.

Communication and Innovation

- Quantum Information Technology and Application Innovation Contest – National First Prize, Runner-up, 2025.
- China International College Students' Innovation Contest – Bronze Medal, 2025.

Mathematical Modeling

- Asia and Pacific Mathematical Contest in Modeling (APMCM) – National First Prize (top 2%), 2025.
- DigitalCup Undergraduate Mathematical Contest in Modeling – National First Prize in Mathematics and Non-Mathematics groups, 2025.
- National Undergraduate Mathematics Competition, Non-Mathematics Category A – Second Prize.

Technical Skills

Programming	Python, C/C++, MATLAB, R, JavaScript/TypeScript, Git, Docker, Linux
Robotics	ROS, OpenCV, PyTorch, CUDA, calibration, kinematics and dynamics, path planning, embedded systems, Arduino/STM32/Raspberry Pi
Perception and ML	3D reconstruction, SfM, visual odometry, object detection, sensor fusion, reinforcement learning, optimization

Languages

Chinese (native) English (TOEFL 110)